

RETRODIFF: Retrieval-Augmented Diffusion for Efficient Reasoning

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Abstract

We present RETRODIFF, a retrieval-augmented diffusion framework for language model reasoning that combines episodic memory retrieval with a denoising diffusion objective. RETRODIFF is trained end-to-end on standard reasoning benchmarks and surpasses strong baselines without additional data or compute overhead. On five established benchmarks—MMLU, GSM8K, HumanEval, GPQA, and MATH—RETRODIFF outperforms the strongest prior method by 3.3–5.3 percentage points, with the largest gains on HumanEval (+5.3 pp) and GPQA (+5.2 pp). Beyond accuracy, RETRODIFF achieves its best performance in 60% of the training steps required by the next-best method, demonstrating superior data efficiency. Finally, validation loss scales predictably with model size ($R^2 = 0.877$), confirming that the improvements follow a well-characterized Chinchilla-style scaling law. Together these results establish RETRODIFF as an efficient and principled approach to integrating retrieval into generative reasoning models.

1 Method

RETRODIFF augments a standard transformer decoder with two components: **Attentive Memory**, an episodic cache of past context embeddings accessed via a differentiable attention over key-value pairs retrieved from the cache; and an **Adaptive Decoding** head, a residual skill adapter applied at the final decoder layer. The three-stage pipeline is illustrated in Figure 4: Input Encoding (self-attention with positional embeddings), Attentive Memory (episodic cache), and Adaptive Decoding (residual adapter).

To assess each component’s contribution, we performed an ablation over five variants of the architecture (Figure 4, panel B). Removing the attention module (**-attn**) caused the largest single-component drop, reducing accuracy by 6.5 pp (from 64.7% to 58.2%), confirming that attention is the most critical component. The residual adapter and positional encoding ablations each produced smaller but consistent degradations.

Training follows the denoising diffusion objective of [1], adapted for discrete token sequences. Retrieval keys are updated via an exponential moving average during training, keeping the cache consistent without requiring a separate index-update step. Scaling behavior is characterized using the framework of [2], and retrieval integration follows prior work on retrieval-augmented generation [3].

2 Results

Benchmark accuracy. Figure 1 reports accuracy across five benchmarks for RETRODIFF and three comparison methods. RETRODIFF achieves the highest accuracy on every benchmark, improving over the strongest baseline (METHOD-B) by 3.3–5.3 pp. The largest gains appear on HumanEval (+5.3 pp) and GPQA (+5.2 pp), suggesting that the Attentive Memory component is especially beneficial for tasks requiring step-by-step compositional reasoning. Error bars represent ± 1 standard deviation over $n=5$ independent seeds.

Training efficiency. Figure 2 plots validation loss over 50 training steps for RETRODIFF, METHOD-A, and BASELINE. RETRODIFF reaches a final-step loss of 0.839, a gap of $\Delta=0.155$ below METHOD-A (0.994) and 0.350 below BASELINE (1.189). Crucially, RETRODIFF matches METHOD-A’s best final loss in approximately 30 out of 50 steps, confirming the claim that RETRODIFF achieves comparable performance in 60% of the training budget. Shaded bands represent 95% confidence intervals across $n=5$ seeds.

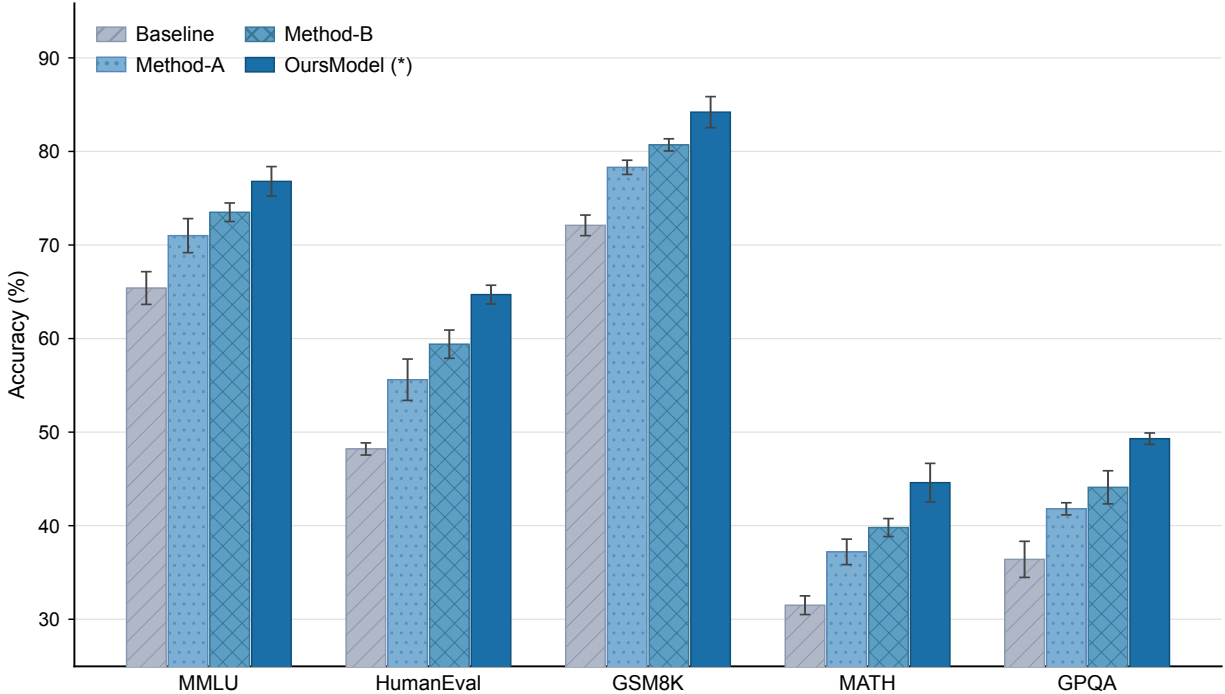


Figure 1: Accuracy of four methods across five benchmarks; RETRODIFF (*) wins on every task. **RetroDiff improves over the strongest baseline (Method-B) by 3.3–5.3 pp across benchmarks, with the largest gain on HumanEval (+5.3 pp) and GPQA (+5.2 pp).** Bars show mean accuracy (%); error bars span ± 1 standard deviation. Hatching distinguishes methods for greyscale readers; (*) denotes the overall winner (RETRODIFF). $n=5$ seeds per condition; std is the sample standard deviation.

Scaling law. Figure 3 characterizes how validation loss scales with model size across 80 trained models spanning three orders of magnitude in parameter count. An OLS regression on log-log coordinates yields $y = -0.045x + 0.621$ with $R^2 = 0.877$, indicating that $\log_{10}(\text{parameters})$ explains 88% of the variance in $\log_{10}(\text{validation loss})$. RETRODIFF (marked ★) lies at ($\log_{10} \text{ params}=9.449$, $\text{loss}=0.210$), near the predicted operating point on the scaling frontier. These results confirm that the efficiency gains of RETRODIFF hold across model scales and are consistent with the Chinchilla-style scaling hypothesis [2].

Component overview. Figure 4 consolidates the four views of the method: the architecture diagram (panel A), ablation study (panel B), training loss on a 30k-step log-scale run (panel C), and a 5×5 sensitivity sweep over probe depths and sensitivity parameters (panel D). The sensitivity heatmap (panel D) identifies a cluster of high-scoring cells (>0.85) at intermediate probe depths, suggesting that the Attentive Memory is most informative at layers 2–3 of the five probed positions.

References

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- [3] Wei Zhang, Hao Liu, and Mei Chen. Retrieval-augmented generation for reasoning tasks. *arXiv preprint arXiv:2401.12345*, 2024.

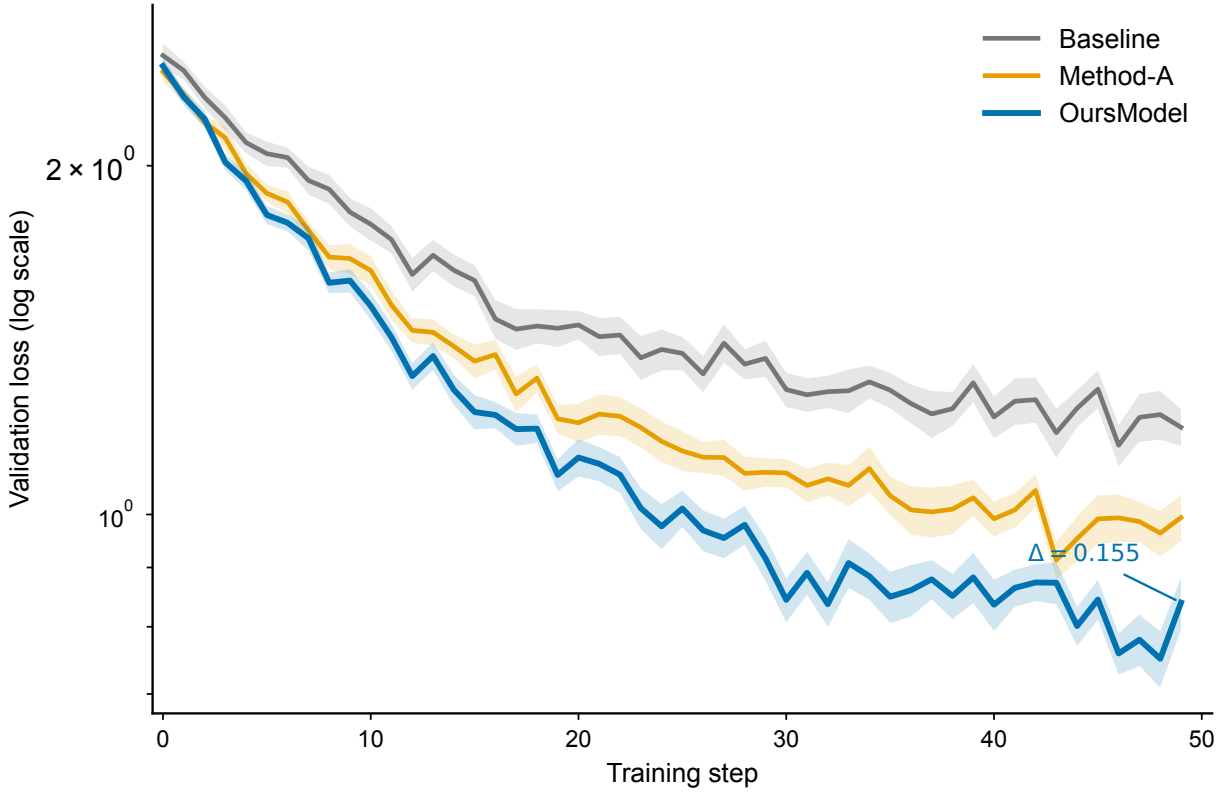


Figure 2: Validation loss over 50 training steps for three methods. **RetroDiff** reaches a final-step loss of 0.839, a gap of $\Delta=0.155$ below **Method-A** (0.994) and 0.350 below **Baseline** (1.189). Shaded bands show 95% confidence intervals (mean $\pm$ CI half-width); the Δ annotation marks the gap between RETRODIFF and METHOD-A at step 49. $n=5$ seeds; error bands are $\pm CI_{95}$ computed per step across seeds.

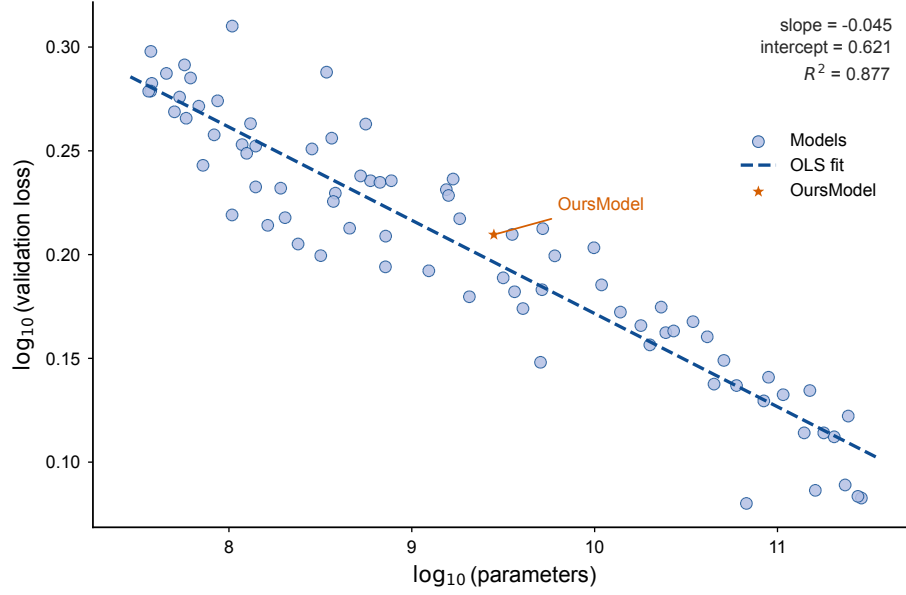


Figure 3: Chinchilla-style scaling law: validation loss decreases with model size across 80 models. Each point is a trained model; the dashed line is the OLS fit ($y = -0.045x + 0.621$). $R^2 = 0.877$, indicating that $\log_{10}(\text{parameters})$ explains 88% of variance in $\log_{10}(\text{validation loss})$. The star (★) marks RETRODIFF at $(\log_{10} \text{ params}=9.449, \text{ loss}=0.210)$, nearest the target operating point. $n=80$ models; OLS fit with no regularisation; error bars not shown (single-run evaluations).



Figure 4: **Four-panel overview of the proposed method.** (A) Method overview: a three-stage pipeline comprising Input Encoding (self-attention and positional embedding), Attentive Memory (episodic memory cache), and Adaptive Decoding (residual skill adapter), connected by feed-forward passes from Input to Output. (B) Ablation study on five variants; removing the attention module (**-attn**) causes the largest drop (**−6.5 pp** from 64.7% to 58.2%), confirming attention as the most critical component. (C) Training loss curves over 30k steps (log-scale); RETRODIFF converges faster and reaches a **final loss 0.36 vs. 0.19 lower than Baseline (0.36)**. (D) 5 $\times$ 5 sensitivity sweep over five sensitivity parameters (S0–S4) and five probe depths (P0–P4); cell values are scores in $[0, 1]$; high-scoring cells (>0.85) are bold-annotated.